

# DriveQueue

## Discrete Event Simulation of a Drive-Thru Service System

Performance Optimization for Queue Waiting Time, Throughput, Bottlenecks, and L  
Sales

### Core Flow

Arrival  
FIFO Queue  
Order  
Payment  
Pickup  
Departure

# Problem Definition

Drive-thru service challenges during normal and peak demand.

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## Main problem

- Customer arrivals fluctuate during the day.
- Service capacity may stay fixed even when demand rises.
- Long queues reduce customer satisfaction.
- A full lane creates rejected vehicles and lost sales.

## Why simulation helps

- Tests scenarios before real changes are made.
- Shows the bottleneck stage clearly.
- Measures waiting time, throughput, and utilization.
- Supports practical staffing and capacity decisions.

# Simulation Objectives

The project measures service performance and compares improvement scenarios.

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## Measure

Average queue waiting time and total time in system.

## Identify

The bottleneck among order, payment, and pickup.

## Evaluate

Throughput, service utilization, and accepted vehicles.

## Compare

Baseline, peak demand, and optimized capacity scenarios.

## Recommend

Practical changes that reduce waiting and congestion.

# Process Diagram

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DriveQueue follows a first-in, first-out drive-thru service flow.



**Lane capacity rule: if the drive-thru lane is full when a vehicle arrives, that vehicle is rejected and counted as a lost sale.**

# Assumptions and Input Data

Input values used for the base simulation runs.

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- Arrivals follow a Poisson-style process.
- Normal arrival rate: 12 vehicles per hour.
- Peak period: 11:00 AM to 2:00 PM.
- Peak demand is modeled with a multiplier.
- Vehicles are served using FIFO queue discipline.

## Service-time assumptions

Order: 1-3 minutes  
Payment: 0.5-1 minute  
Pickup: 1-2 minutes  
Baseline capacity: 1 station/window per stage  
Lane capacity: 10 vehicles

# Implementation Details

DriveQueue combines a SimPy model with a Tkinter visualization interface.

**SimPy**

Simulation engine

**Tkinter**

Desktop interface

**FIFO**

Queue rule

**KPIs**

Measured output

The interface allows the user to adjust duration, arrival rate, peak multiplier, service-window counts, lane capacity, and random seed. It also shows animated vehicle movement, event logs, performance metrics, bottleneck detection, and scenario recommendations.



# Experimental Results

Scenario comparison using served vehicles, rejected vehicles, waiting time, and system time.

| Scenario       | Served | Rejected | TP/hr | Avg Wait | Avg System | Acceptance |
|----------------|--------|----------|-------|----------|------------|------------|
| Baseline       | 297    | 2        | 12.4  | 2.02 min | 6.28 min   | 99.3%      |
| Peak demand    | 344    | 23       | 14.3  | 2.86 min | 7.12 min   | 93.8%      |
| Optimized cap. | 345    | 0        | 14.4  | 0.29 min | 4.51 min   | 100.0%     |

**Main reading: optimization keeps throughput high while removing rejected vehicles and sharply reducing waiting time.**

# Detailed Key Findings

The results show where the drive-thru improves and where congestion appears.

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## Bottleneck

Order taking remains the main bottleneck because it starts the service chain and affects all following stages.

## Lost sales

Peak demand creates 23 rejected vehicles, showing that a full lane can directly reduce possible sales.

## Optimization

Adding capacity eliminates rejections and lowers average queue wait to 0.29 minutes.

## Customer flow

Average total system time improves from 7.12 minutes during peak demand to 4.51 minutes after optimization.



# Queue Wait Time Comparison

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Average queue wait is the clearest customer-experience indicator.



## Interpretation

Peak demand raises waiting time by about 42% versus baseline. The optimized setup reduces peak waiting by about 90%, which is the strongest evidence that added capacity improves service flow.

# Conclusions and Recommendations

DriveQueue shows how simulation supports real service-system decisions.

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## Conclusion

- The model identifies bottlenecks before real changes are made.
- Peak demand causes longer waits and more rejected vehicles.
- Optimized capacity gives the best balance of throughput and customer experience.

## Recommendation

- Improve the order stage during peak hours.
- Use added staff or faster order-taking tools.
- Support pickup coordination when demand is high.
- Monitor lost vehicles as a warning sign of lane congestion.

# Thank You

DriveQueue: Drive-Thru Service System Performance Optimization

Questions and discussion